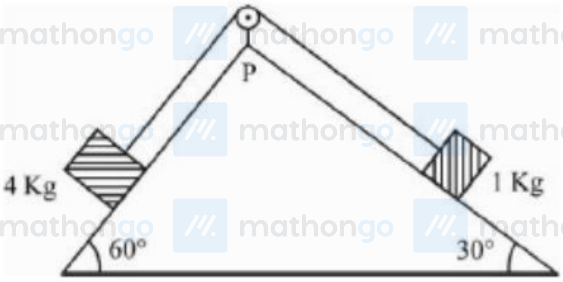
Q1 - 24 January - Shift 1

As per given figure, a weightless pulley P is attached on a double inclined frictionless surface. The tension in the string (massless) will be (if g = 10 m/s^2)



(1) $(4\sqrt{3}+1)$N (2) $4(\sqrt{3}+1)$N

(3) $4(\sqrt{3}-1)$N (4) $(4\sqrt{3}-1)$N

Space for your notes:

Q2 - 24 January - Shift 1

Given below are two statements :

Statement-I : An elevator can go up or down with uniform speed when its weight is balanced with the tension of its cable.

Statement-II : Force exerted by the floor of an elevator on the foot of a person standing on it is more than his/her weight when the elevator goes down with increasing speed.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both statement I and statement II are false
(2) Statement I is true but Statement II is false
(3) Both Statement I and Statement II are true
(4) Statement I is false but Statement II is true

Space for your notes:

Q3 - 25 January - Shift 2

Consider a block kept on an inclined plane (inclined at 45°) as shown in the figure. If the force required to just push it up the incline is 2 times the force required to just prevent it from sliding down, the coefficient of friction between the block and inclined plane (μ) is equal to :

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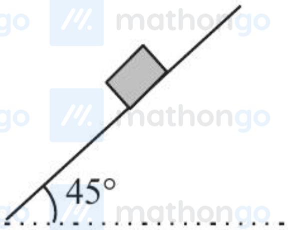


(1) 0.33 (2) 0.60
(3) 0.25 (4) 0.50

Q4 - 29 January - Shift 1

A block of mass m slides down the plane inclined at angle 30° with an acceleration $\frac{g}{4}$. The value of coefficient of kinetic friction will be :

Space for your notes:

(1) $\frac{2\sqrt{3}+1}{2}$ (2) $\frac{1}{2\sqrt{3}}$
(3) $\frac{\sqrt{3}}{2}$ (4) $\frac{2\sqrt{3}-1}{2}$

Q5 - 29 January - Shift 2

Force acts for 20 s on a body of mass 20 kg, starting from rest, after which the force ceases and then body describes 50 m in the next 10 s. The value of force will be :

Space for your notes:

(1) 40 N (2) 5 N
(3) 20 N (4) 10 N

Q6 - 29 January - Shift 2

The time taken by an object to slide down 45° rough inclined plane is n times as it takes to slide down a perfectly smooth 45° incline plane. The coefficient of kinetic friction between the object and the incline plane is

(1) $\sqrt{\frac{1}{1-n^2}}$ (2) $\sqrt{1-\frac{1}{n^2}}$

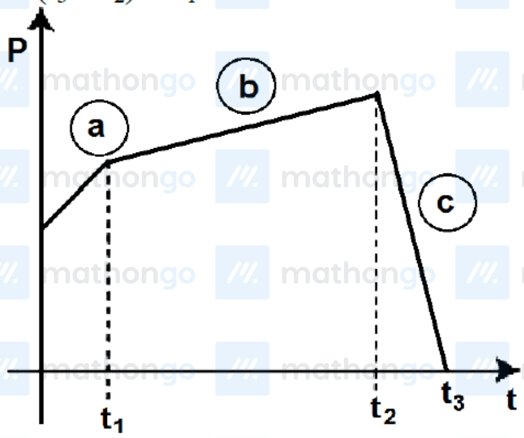
(3) $1+\frac{1}{n^2}$ (4) $1-\frac{1}{n^2}$

Space for your notes:

Q7 - 30 January - Shift 1

The figure represents the momentum time (p-t) curve for a particle moving along an axis under the influence of the force. Identify the regions on the graph where the magnitude of the force is maximum and minimum respectively ?

If $(t_3 - t_2) < t_1$.

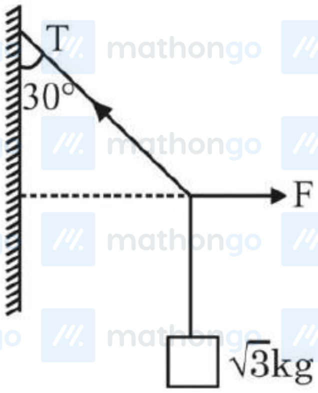


(1) c and a (2) b and c
(3) c and b (4) a and b

Space for your notes:

Q8 - 30 January - Shift 2

A block of $\sqrt{3}$ kg is attached to a string whose other end is attached to the wall. An unknown force F is applied so that the string makes an angle of 30° with the wall. The tension T is :
(Given $g = 10\ ms^{-2}$)



(1) 20 N
(2) 25 N
(3) 10 N
(4) 15 N

Space for your notes:

Q9 - 30 January - Shift 2

A stone tied to 180 cm long string at its end is making 28 revolutions in horizontal circle in every minute. The magnitude of acceleration of stone is $\frac{1936}{x}\ ms^{-2}$. The value of x ________.

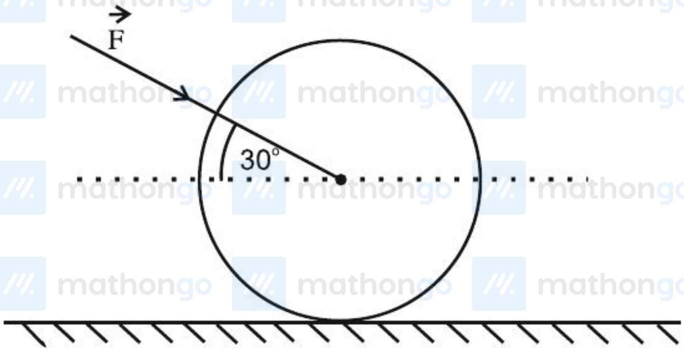
$\left(\text{Take } \pi = \frac{22}{7}\right)$

Space for your notes:

Q10 - 31 January - Shift 1

As shown in figure, a 70 kg garden roller is pushed with a force of $\vec{F} = 200\,N$ at an angle of 30^{o} with horizontal. The normal reaction on the roller is (Given $g = 10\ m\ s^{-2}$)

Space for your notes:



(1) $800\sqrt{2}\,N$

(2) $600\,N$

(3) $800\,N$

(4) $200\sqrt{3}\,N$

Q11 - 31 January - Shift 1

A lift of mass M = 500 kg is descending with speed of 2 ms^{-1}. Its supporting cable begins to slip thus allowing it to fall with a constant acceleration of 2 ms^{-2}. The kinetic energy of the lift at the end of fall through to a distance of 6 m will be ____ kJ.

Space for your notes:

Q12 - 31 January - Shift 2

A stone of mass 1 kg is tied to end of a massless string of length 1 m. If the breaking tension of the string is 400 N, then maximum linear velocity, the stone can have without breaking the string, while rotating in horizontal plane, is:

(1) 20 ms^{-1}
(2) 40 ms^{-1}
(3) 400 ms^{-1}
(4) 10 ms^{-1}

Space for your notes:

Q13 - 31 January - Shift 2

A body of mass 10 kg is moving with an initial speed of 20 m/s. The body stops after 5 s due to friction between body and the floor. The value of the coefficient of friction is: (Take acceleration due to gravity $g = 10\ ms^{-2}$)

(1) 0.2 (2) 0.3
(3) 0.5 (4) 0.4

Space for your notes:

Q14 - 01 February - Shift 1

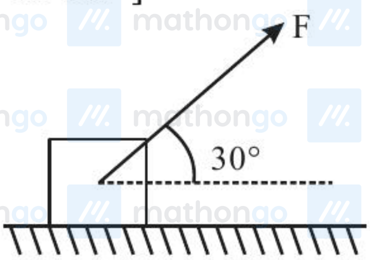
A block of mass 5 kg is placed at rest on a table of rough surface. Now, if a force of 30N is applied in the direction parallel to surface of the table, the block slides through a distance of 50 m in an interval of time 10s. Coefficient of kinetic friction is (given, $g = 10\ ms^{-2}$):

(1) 0.60 (2) 0.75
(3) 0.50 (4) 0.25

Space for your notes:

Q15 - 01 February - Shift 2

As shown in the figure a block of mass 10 kg lying on a horizontal surface is pulled by a force F acting at an angle 30°, with horizontal. For $\mu_s = 0.25$, the block will just start to move for the value of F: [Given $g = 10\ ms^{-2}$]



(1) 33.3 N (2) 25.2 N
(3) 20 N (4) 35.7 N

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2) **Q2** (2) **Q3** (1) **Q4** (2)

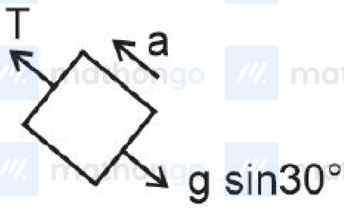
Q5 (2) **Q6** (4) **Q7** (3) **Q8** (1)

Q9 (125) **Q10** (3) **Q11** (7) **Q12** (1)

Q13 (4) **Q14** (3) **Q15** (2)

Q1 (2)

$$4g\sin 60^\circ - T = 4a \qquad \ldots(1)$$



$$T - g\sin 30^\circ = a \qquad \ldots(2)$$

Solving (1) and (2) we get.

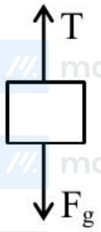
$$20\sqrt{3} - T = 4T - 20$$

$$T = 4(\sqrt{3}+1)\text{N}$$

Q2 (2)

Statement-1

When elevator is moving with uniform speed $T = F_g$



Statement-2

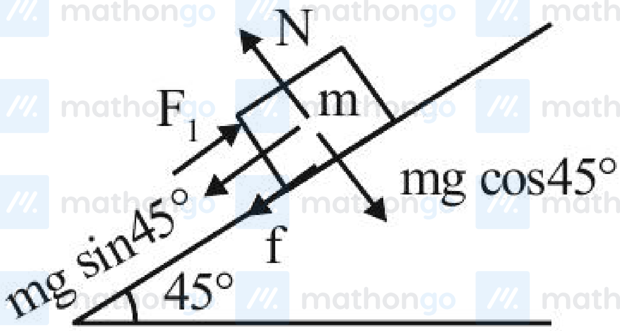
When elevator is going down with increasing speed, its acceleration is downward.

Hence

$$W - N = \frac{W}{g} \times a$$

$$N = W\left(1 - \frac{a}{g}\right)$$ i.e. less than weight.

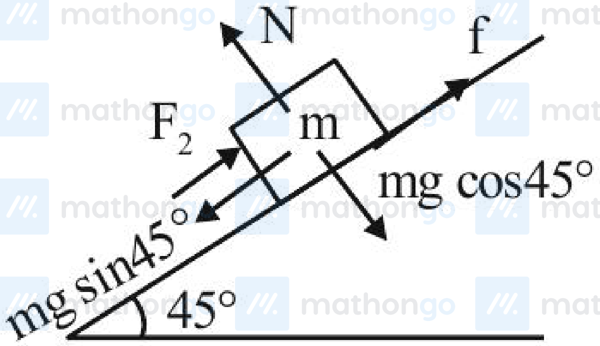
Q3 (1)



$F_1 = mg\sin 45° + f = mg\sin 45° + \mu N$

$F_1 = \frac{mg}{\sqrt{2}} + \mu mg\cos 45°$

$F_1 = \frac{mg}{\sqrt{2}}(1+\mu)$



$F_2 = mg\sin 45° - f = mg\sin 45° - \mu N$

$= \frac{mg}{\sqrt{2}}(1-\mu)$

$F_1 = 2F_2$

$\frac{mg}{\sqrt{2}}(1+\mu) = 2\frac{mg}{\sqrt{2}}(1-\mu)$

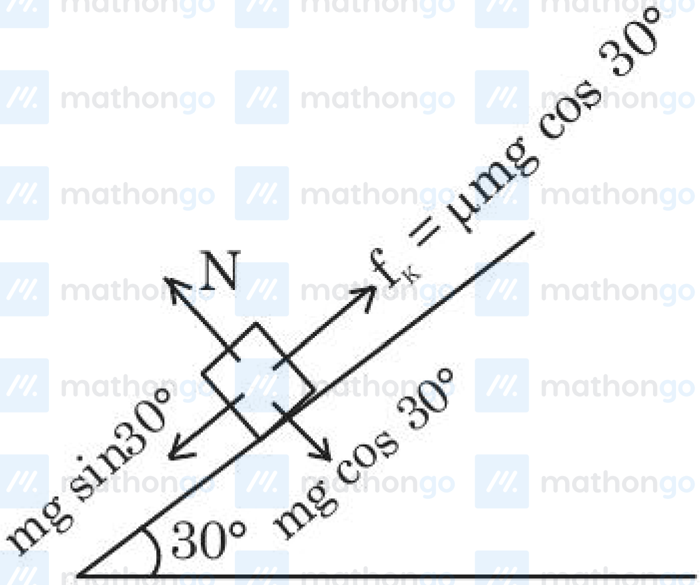
$1 + \mu = 2 - 2\mu$

$\mu = 1/3 = 0.33$

Q4 (2)

$$Mg \sin 30° - \mu mg\cos 30° = ma$$

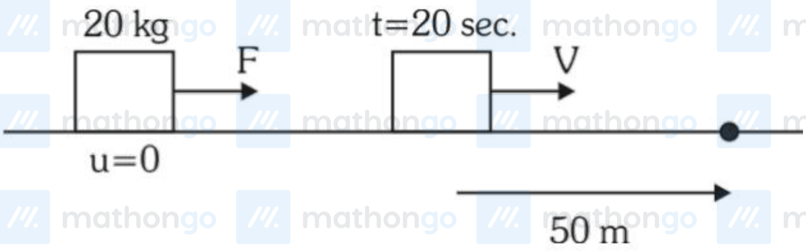
$$\frac{g}{2} - \frac{\sqrt{3}}{2}.\mu g = \frac{g}{4}$$



$$\frac{\sqrt{3}}{2}\mu = \frac{1}{4}$$

$$\mu = \frac{1}{2\sqrt{3}}$$

Q5 (2)



$50 = V \times 10$

$V = 5 \text{ m/s}$

$V = 0 + a \times 20$

$5 = a \times 20$

$a = \frac{1}{4} \text{ m/s}^2$

$F = ma = 20 \times \frac{1}{4} = 5N$

Q6 (4)

$$a_1 = g \sin\theta = g/\sqrt{2}$$

$$a_2 = g\sin\theta - Kg\cos\theta = \frac{g}{\sqrt{2}} - \frac{Kg}{\sqrt{2}}$$

$$t_2 = nt_1 \quad \& \quad a_1 t_1^2 = a_2 t_2^2$$

$$\frac{g}{\sqrt{2}} t_1^2 = \left(\frac{g}{\sqrt{2}} - \frac{kg}{\sqrt{2}}\right) n^2 t_1^2$$

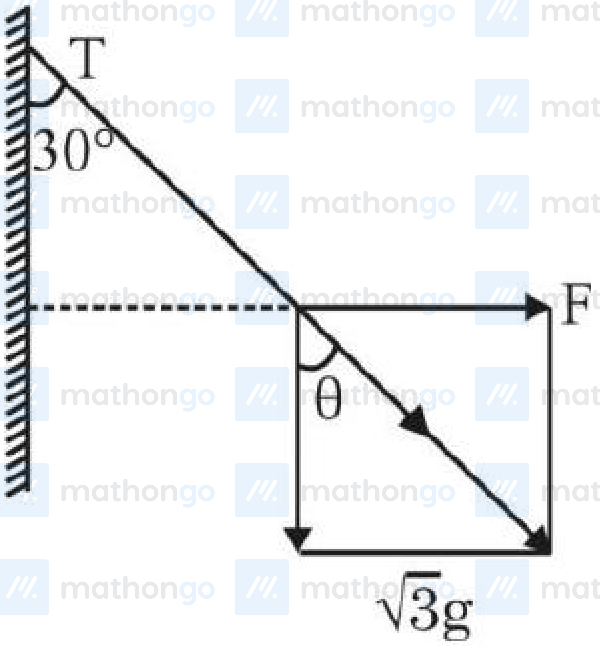
$$K = 1 - \frac{1}{n^2} \quad \textbf{Ans. 4}$$

Q7 (3)

$$\left|\frac{d\vec{p}}{dt}\right| = |\vec{F}| \quad \Rightarrow \quad \frac{d\vec{p}}{dt} = \text{Slope of curve}$$

max slope (c)
min slope (b)

Q8 (1)



$\theta = 30°$

$$\cos\theta = \frac{\sqrt{3}g}{T}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{\sqrt{3}g}{T}$$

$$\Rightarrow T = 20N$$

Q9 (125)

$$a = \omega^2 R = \left(\frac{28 \times 2\pi}{60}\right)^2 \times 1.8$$

$$= \left(\frac{56}{60} \times \frac{22}{7}\right)^2 \times 1.8$$

$$= \frac{(44)^2}{225} \times 1.8$$

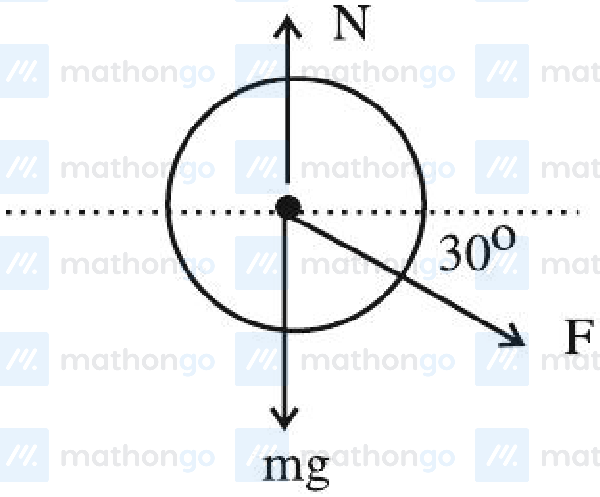
$$= \frac{1936 \times 1.8}{225}$$

$$x = 125$$

Q10 (3)

$$N = mg + F \sin 30^\circ$$

$$= 700 + 200 \times \frac{1}{2} = 800 \text{ newton.}$$



Q11 (7)

$v^2 = u^2 + 2as$

$= 2^2 + 2(2)(6)$

$= 4 + 24 = 28$

$KE = \frac{1}{2}mv^2$

$= \frac{1}{2}(500)28$

$= 7000 \text{ J}$

$= 7 \text{ kJ}$

$\boxed{\text{Ans. } 7}$

Q12 (1)

No Solution

Q13 (4)

No Solution

Q14 (3)

$$S = ut + \frac{1}{2}at^2$$

$$50 = 0 + \frac{1}{2} \times a \times 100$$

$$a = 1 \text{ m/s}^2$$

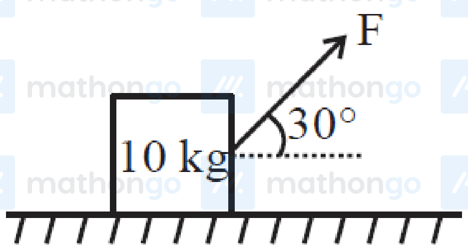
$$F - \mu mg = ma$$

$$30 - \mu \times 50 = 5 \times 1$$

$$50\mu = 25$$

$$\mu = \frac{1}{2}$$

Q15 (2)



$$N = Mg - F\ \text{Sin}\ 30^\circ$$

$$= mg - \frac{F}{2} = 100 - \frac{F}{2} = \frac{200 - F}{2}$$

$$F\ \text{Cos}\ 30^\circ = \mu N$$

$$\sqrt{3}\frac{F}{2} = 0.25 \times \left(\frac{200 - F}{2}\right)$$

$$4\sqrt{3}F = 200 - F$$

$$F = \frac{200}{4\sqrt{3} + 1} = 25.22$$